Retail Sales Data Analytics & Business Intelligence

InternNova Week 6 Capstone | Atanu Manna

Project Overview

This capstone analyzes real transaction-level data from the UCI Online Retail dataset. The workflow includes data inspection, cleaning, exploratory analysis, visualizations, Power BI reporting, business recommendations and Git/GitHub version control.

Dataset and Attribution

Source: UCI Machine Learning Repository — Online Retail. The dataset covers transactions from a UK-based online retailer from 1 December 2010 to 9 December 2011. Citation: Chen, D. (2015). *Online Retail* [Dataset]. UCI Machine Learning Repository. DOI: 10.24432/C5BW33. License: CC BY 4.0.

Official source: https://archive.ics.uci.edu/dataset/352/online+retail

Data Preparation Method

The analysis preserves the raw file and creates a separate processed dataset. It standardizes text fields, parses dates and numeric columns, removes exact duplicate rows, flags cancellation invoices (invoice numbers beginning with C), and calculates line revenue as Quantity × UnitPrice in GBP. Rows with cancellations, non-positive quantity, non-positive price, invalid dates or missing descriptions are excluded from positive-sales analysis. Missing customer identifiers are not filled with invented values.

Exploratory Data Analysis

The analysis script generates descriptive statistics, a numeric correlation matrix, monthly and country-level summaries, product rankings, customer summaries and an outlier audit using the 1.5×IQR rule. Outliers are flagged rather than automatically deleted. The numeric results should be copied from the generated files after running the pipeline on the official dataset.

Visualizations

Six visualizations are generated by the script: monthly revenue trend, top countries excluding the UK, top products, quantity versus line revenue, correlation heatmap, and revenue by weekday. Insert the actual generated charts from the visuals folder and explain the observed trends.

Power BI Dashboard

Build and save a native Power BI Desktop report using data/processed/cleaned_sales.csv. Recommended KPI cards are Total Revenue (£), Total Invoices and Unique Customers. Include at least four visuals: monthly revenue line chart, top-products bar chart, country revenue bar chart, and weekday revenue column chart. Add slicers for Country, InvoiceDate and product. Save the file as powerbi/Retail_Sales_Dashboard.pbix and insert genuine screenshots.

Business Insights and Recommendations

Use actual outputs from analysis_audit.json, monthly_sales.csv, country_sales.csv and product_sales.csv to write at least five findings. Recommendations should be tied directly to the highest/lowest-performing products, countries, and months. Revenue is not profit because the source dataset has no cost information.

GitHub and Version Control

Create a public repository named retail-sales-analytics. Commit the project structure, then the cleaning/EDA pipeline, then the analysis outputs and documentation. Include at least three meaningful commits and verify the repository URL before submission.

Conclusion and Limitations

This project provides a reproducible workflow for retail transaction analysis. Final numerical claims and business recommendations must be based on the actual run outputs. The dataset represents historical transactions and should not be treated as current market performance. This draft is not proof that the Power BI report or GitHub pushes have been completed.

Completion checklist

- ■ Download official dataset and run analysis script
- ■ Review actual metrics and insert charts
- ■ Build and save real .pbix dashboard
- ■ Capture genuine Power BI screenshots
- ■ Complete insights and recommendations
- ■ Make at least three Git commits and push to GitHub
- ■ Replace this draft with verified final report